A Completely Isometric Calculus Without Similarity to an m-Hypercontraction

Run fa_banach_001, agent agent_lane_11

17 August 2026

Source Question and Answer

Popescu proves that similarity to a tuple in a radial noncommutative variety of order m implies complete boundedness of its polynomial calculus on the variety algebra. He proves the converse for $m = 1$ and states that it is open for $m \geq 2$ [1, Introduction and Section 3]. The nearby pure version also assumes $\Phi_A^k(I) \to 0$ strongly.

We give a negative answer for every $m \geq 2$, already in one variable with no constraints. The calculus in the example is not merely completely bounded: it is completely isometric. The strong-purity condition also holds. Status: *candidate full counterexample, likely valid*, subject to human review.

The Two Model Shifts

Fix an integer $m \geq 2$, take one variable, $f = X$, and take the homogeneous constraint set $\mathcal{P} = \{0\}$. The coefficients of the source's universal model are

$$b_k^{(m)} = \binom{k+m-1}{m-1} \qquad (k \geq 0), \tag{1}$$

because the word of length k has only the decomposition into k words of length one with nonzero f-coefficients. Hence the universal weighted shift W_m on $\ell^2(\mathbb{N}_0)$ is

$$W_m e_k = \sqrt{\frac{b_k^{(m)}}{b_{k+1}^{(m)}}}\, e_{k+1} = \sqrt{\frac{k+1}{k+m}}\, e_{k+1}. \tag{2}$$

Let $\mathcal{H}_m$ be the reproducing-kernel Hilbert space on $\mathbb{D}$ with kernel

$$K_m(z,w) = \frac{1}{(1-z\overline{w})^m} = \sum_{k\geq 0} b_k^{(m)} z^k \overline{w}^k. \tag{3}$$

The unitary $e_k \mapsto \sqrt{b_k^{(m)}}z^k$ identifies W_m with multiplication by z on $\mathcal{H}_m$. For $m \geq 2$, this is the weighted Bergman space with norm, up to an irrelevant positive constant,

$$\|h\|_m^2 = \int_{\mathbb{D}} |h(z)|^2 (1-|z|^2)^{m-2}\, dA(z). \tag{4}$$

Lemma 1. *For every $d \geq 1$ and every $d \times d$ matrix polynomial P,*

$$\|P(W_m)\| = \sup_{z\in\mathbb{D}} \|P(z)\|. \tag{5}$$

The same identity holds with W_m replaced by the unilateral shift S.

Proof. The pointwise estimate in (4), applied to vector-valued functions, gives the upper bound in (5). If k_z is the normalized reproducing kernel, then

$$P(W_m)^*(k_z \otimes v) = k_z \otimes P(z)^* v,$$

so the reverse inequality follows by taking unit vectors v and then the supremum over z. For $S = M_z$ on Hardy space, the identical argument uses the boundary L^2 norm and the Hardy reproducing kernels. ☐

It follows that

$$\Psi : \mathcal{A}_1(\mathcal{V}^m_{X,\{0\}}) \longrightarrow B(H^2), \qquad \Psi(p(W_m)) = p(S), \tag{6}$$

extends from polynomials to a unital complete isometry.

The Similarity Obstruction

Theorem 1. *For every integer $m \geq 2$, the homomorphism (6) is completely isometric, but S is not similar to any operator in $\mathcal{V}^m_{X,\{0\}} = \mathbf{D}^m_X$. Moreover,*

$$\Phi^k_{X,S}(I) = S^k S^{*k} \longrightarrow 0 \quad \textit{strongly.} \tag{7}$$

Consequently, complete boundedness of the variety-algebra calculus does not imply the similarity asserted in the source's open converse for any $m \geq 2$, even with its strong-purity side condition.

Proof. Complete isometry is Lemma 1. Suppose, toward a contradiction, that

$$S = Y^{-1} T Y, \qquad T \in \mathbf{D}^m_X, \tag{8}$$

for an invertible Y. Since $m \geq 2$, the definition of the domain gives

$$(I - \Phi_{X,T})^2(I) = I - 2TT^* + T^2 T^{*2} \geq 0. \tag{9}$$

Set $R = Y^{-1} Y^{-*}$. Congruencing (9) by Y^{-1} and Y^{-*} and using $T = YSY^{-1}$ yields

$$R - 2SRS^* + S^2 R S^{*2} \geq 0. \tag{10}$$

Write $r_k = \langle Re_k, e_k \rangle$. The e_1 diagonal entry of (10) is

$$r_1 - 2r_0 \geq 0, \tag{11}$$

because $S^* e_1 = e_0$ and $S^{*2} e_1 = 0$. For every $k \geq 2$, the e_k entry is

$$r_k - 2r_{k-1} + r_{k-2} \geq 0. \tag{12}$$

Let $d_k = r_k - r_{k-1}$. Equations (11) and (12) say

$$d_1 \geq r_0, \qquad d_k \geq d_{k-1} \quad (k \geq 2).$$

Since R is positive and invertible, $R \geq cI$ for some $c > 0$, so $r_0 \geq c$. Therefore

$$r_k = r_0 + \sum_{j=1}^{k} d_j \geq r_0 + k d_1 \geq (k+1)c, \tag{13}$$

contradicting $r_k \leq \|R\|$. Thus the similarity (8) cannot exist.

Finally, $S^k S^{*k}$ is the projection onto $\overline{\operatorname{span}}\{e_j : j \geq k\}$, and these tail projections converge strongly to zero. This proves (7). ☐

Why the Argument Answers the Exact Question

The specialization $f = X$, $\mathcal{P} = \{0\}$ is allowed in the source; the domain is radial and the constraints are homogeneous. The example is on a Hilbert space, as in the stated formulation. It defeats the converse at its strongest input level (complete isometry) and retains the nearby purity hypothesis. Only the second defect is needed, so the same operator works simultaneously for all orders $m \geq 2$.

The obstruction is insensitive to off-diagonal entries of R: positivity of (10) forces the displayed scalar diagonal inequalities. The boundary equation (11) then supplies a strictly positive initial slope, which convexity propagates into forbidden linear growth.

Novelty Check and Verification

Cheap indexes and bounded searches through 17 August 2026 by exact wording, title, arXiv id, m-hypercontractions, completely bounded calculi, Bergman shifts, and later similarity work found no resolution of this exact converse. The later polydomain paper [2] gives cone criteria but does not answer it. This is not a priority claim.

The companion script checks (1)–(2), the unilateral shift's second-defect diagonal, the extremal recurrence behind (13), and strong purity on finite-support vectors. These are auxiliary transcription checks; the proof above is exact.

Human Review Recommendation

First verify the complete matrix norm identity (5). Then audit the congruence producing (10), including the placement of Y^{-1} and Y^{-*}. The remaining diagonal convexity argument is elementary and closes the counterexample.

References

[1] G. Popescu, *Joint similarity to operators in noncommutative varieties*, Proc. London Math. Soc. (3) **103** (2011), 331–370; arXiv:1004.2021v2.

[2] G. Popescu, *Similarity problems in noncommutative polydomains*, arXiv:1412.1672.